

FINAL - Due on March 15, 2025

1. Implement a method of line using central difference on space and 2nd order RK to solve

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + v(x) - cu$$

in $(0, 1)$ with boundary conditions: $\frac{\partial u(x,t)}{\partial x} = 0$ at $x = 0$ and $u(x, t) = 0$ at $x = 1$. $v(x) = v_0$ for $x \in (0, 0.1)$ and $v(x) = 0$ otherwise. v_0 , D , c are constants. Demonstrate order of accuracy for the implementation. Present simulation results for cases that achieve steady state solutions.

Please test the accuracy of the method and study the steady-state solutions using different initial conditions.

2. Please study the paper:

Developmental Cell, Vol. 2, 785–796, June, 2002, Copyright ©2002 by Cell Press

Do Morphogen Gradients Arise by Diffusion?

Then

- 1) Develop and implement a code, which is NOT needed to be the same as described in the section “Numerical Solutions”, to reproduce Figure 4 and Figure 6.
 - 2) Please elaborate how you test your code that should be easily adapted for solving various one-spatial-dimension morphogen systems that incorporate more variables and equations than those in the paper.
 - 3) Reproduce Figure 5.
3. Please describe in words or figures *without any mathematical equations* on what you’ve learned in this class. Please keep the presentation within three pages. This problem will be graded based on a) the clarity of overall logic, b) description of connections among different topics, c) discussions on strengths and weaknesses of different methods, and d) speculations on potential systems biology applications of the studied methods.